

SQL + Spring Boot Roadmap

Stage 1 - SQL Foundations

- RDBMS Concepts
- Tables
- Primary & Foreign Keys
- Normalization

Stage 2 - Querying Data

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- HAVING

Stage 3 - Joins & Advanced SQL

- Inner Join
- Outer Joins
- Subqueries
- CTE
- Views

Stage 4 - Database Performance

- Indexes
- Execution Plans
- Optimization Basics

Stage 5 - JDBC

- Connections
- Prepared Statements
- Connection Pooling

Stage 6 - Spring Boot Basics

- Starters
- Auto Configuration
- Profiles
- Actuator

Stage 7 - Spring Data JPA

- Repositories
- CRUD Operations
- Paging & Sorting

Stage 8 - Hibernate

- Entity Mapping
- Relationships
- Fetch Types

Stage 9 - Transactions

- @Transactional
- ACID Concepts

Stage 10 - Advanced Persistence

- JPQL
- Native Queries
- Specifications

Stage 11 - Database Migration

- Flyway
- Liquibase

Stage 12 - Testing

- H2 Database
- Integration Testing

Stage 13 - Deployment

- MySQL/PostgreSQL Integration
- Docker

Stage 14 - Interview Preparation

- SQL Interview Questions
- Spring Data JPA Questions
- Scenario-Based Problems